NP-hardness of approximating sublinear-time conditional Kolmogorov complexity via secret sharing

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Editorial note. Faithful editable transcription of formal author Theorem6.6 and complete Section6.2 main proof, including the full generic secret-sharing reduction Theorem6.14, all Claims6.17-6.23, exact cited prior premises and every page25-29 continuation. The generic reduction and privacy support are uncounted context. Original mathematical expressions and CC BY4.0 license retained; no new proof or correction.

Allowed standard prerequisites (editorial).

- Efficient secret sharing for monotone access structures and MMSA gap hardness. Exact cited Benaloh-Leichter Lemma6.11 and Dinur-Safra/Dinur et al. Lemma6.13, including original author Proof Sketch, retained.
- Amortized time-bounded Kolmogorov complexity approaches entropy. Exact prior Lemma6.15 citing [6,5] retained; author computational-depth corollary supplied in full.

3. Preliminaries: Notation. $[n]$ denotes $\{1, \dots, n\}$. For a function $p : \mathbb{N} \to \mathbb{N}$ and $i \in \mathbb{N}$, the function $p^{(i)} : \mathbb{N} \to \mathbb{N}$ is recursively defined as follows: $p^{(0)}(n) := n$ and $p^{(i+1)}(n) := p^{(i)}(p(n))$ for every $n \in \mathbb{N}$.

Kolmogorov Complexity. For a string $x \in \{0,1\}^*$ and an index $i \in \mathbb{N}$, let x_i denote the i-th bit of x if $i \leq |x|$ and $\perp$ if $i > |x|$. We fix an efficient universal Turing machine U. Time-bounded Kolmogorov complexity is formally defined as follows.

Definition 3.1 (Time-bounded Kolmogorov complexity). For strings $x, y \in \{0,1\}^*$, a time bound $t \in \mathbb{N} \cup \{\infty\}$, and an oracle A, the A-oracle t-time-bounded Kolmogorov complexity of x given y is defined as

$$K^{t,A}(x \mid y) := \min\{|d| : U^{A,y,d} \text{ outputs } x_i \text{ on input } i \text{ in time } t \text{ for every } i \in [1, \dots, |x|+1]\}.$$

Here, $U^{A,y,d}$ indicates that the universal Turing machine is given oracle access to A and each bit of y and $d \in \{0,1\}^*$. We omit the superscript A if $A = \varnothing$, the superscript t if $t = \infty$, and "$\mid y$" if y is the empty string. A simple counting argument implies the following basic fact of Kolmogorov-randomness.

Fact 3.2. For any integer $s \geq 1$ and any string $y \in \{0,1\}^*$, the number of strings $x \in \{0,1\}^*$ such that $K(x \mid y) < s$ is less than 2^s.

Proof. The number of programs of length less than s is at most $\sum_{i=0}^{s-1} 2^i < 2^s$. □

Definition 3.5 (Time-Bounded Computational Depth [9,35]). For time bounds $s \in \mathbb{N}$ and $t \in \mathbb{N} \cup \{\infty\}$ and a string $x \in \{0,1\}^*$, the (s,t)-time-bounded computational depth of x is defined as

$$\mathrm{cd}^{s,t}(x) := K^s(x) - K^t(x).$$

We omit the superscript t if $t = \infty$.

Definition 6.1. For a polynomial $\tau : \mathbb{N}^3 \to \mathbb{N}$, we define the promise problem $\mathsf{Gap}_\tau\mathsf{MINcKT}$ to be $(\Pi_{\mathrm{Yes}}, \Pi_{\mathrm{No}})$ such that

$$\Pi_{\mathrm{Yes}} := \{(x, y, 1^t, 1^s) : K^t(x \mid y) \leq s - \mathrm{cd}^{t,\tau(|x|,|y|,t)}(y)\},$$

$$\Pi_{\mathrm{No}} := \{(x, y, 1^t, 1^s) : K^{\tau(|x|,|y|,t)}(x \mid y) > s + \log \tau(|x|, |y|, t)\}.$$

Definition 6.5. For a polynomial $\tau : \mathbb{N}^3 \to \mathbb{N}$ and a function $\sigma : \mathbb{N} \to \mathbb{N}$, we define the promise problem $\mathsf{Gap}_{\tau,\sigma}\mathsf{MINcKT}$ to be $(\Pi_{\mathrm{Yes}}, \Pi_{\mathrm{No}})$ such that

$$\Pi_{\mathrm{Yes}} := \{(x, y, 1^t, 1^s) : K^t(x \mid y) \leq s - \mathrm{cd}^{t,\tau(|x|,|y|,t)}(y)\},$$

$$\Pi_{\mathrm{No}} := \{(x, y, 1^t, 1^s) : K^{\tau(|x|,|y|,t)}(x \mid y) > \sigma(|x|) \cdot s + \log \tau(|x|, |y|, t)\}.$$

Note that $\mathsf{Gap}_{\tau,1}\mathsf{MINcKT} = \mathsf{Gap}_\tau\mathsf{MINcKT}$ is trivially reducible to $\mathsf{Gap}_{\tau,\sigma}\mathsf{MINcKT}$ for every $\sigma \geq 1$.

Definition 6.7 (Access Structure). An access structure $\mathcal{A} \subseteq 2^{[n]}$ is a "monotone" collection of subsets of $[n]$; that is, for every $T \supseteq S \in \mathcal{A}$, we have $T \in \mathcal{A}$. The minimum weight of $\mathcal{A}$ is defined to be $w(\mathcal{A}) := \min\{|T| : T \in \mathcal{A}\}$. We will prove that there is a generic reduction from the problem of estimating the weight of access structures $\mathcal{A}$ to $\mathsf{GapMINcKT}$ if there exists an efficient secret sharing scheme for $\mathcal{A}$.

Definition 6.8 (Secret Sharing [12]). A secret sharing scheme for $\mathcal{A}$ is a pair $(\mathsf{Share}, \mathsf{Rec})$ of a randomized algorithm Share and a deterministic algorithm Rec with the following properties for every $\ell \in \mathbb{N}$:

1. Correctness: For every $T \in \mathcal{A}$ and for every string $x \in \{0,1\}^\ell$, any output of $\mathsf{Share}(x)$ is a sequence $(y_1, \ldots, y_n)$ of n strings that satisfies $\mathsf{Rec}(y_T) = x$, where $y_T := \{(i, y_i) : i \in T\}$.

2. Privacy: For every $T \notin \mathcal{A}$ and for every random variable X on $\{0,1\}^\ell$, the random variables X and $\mathsf{Share}(X)_T$ are statistically independent.

We observe that the privacy condition can be stated in terms of Kolmogorov complexity. We mention in passing that Kolmogorov complexity-theoretic versions of privacy conditions are studied in, e.g., [10,48].

Definition 6.10. A family $\mathcal{A} = \{\mathcal{A}_\phi\}_{\phi\in\{0,1\}^*}$ of access structures is said to admit efficient secret sharing schemes if there exists a pair $(\mathsf{Share}, \mathsf{Rec})$ of a randomized polynomial-time algorithm Share and a deterministic polynomial-time algorithm Rec such that for every $\phi \in \{0,1\}^*$, the pair $(\mathsf{Share}(\phi,-), \mathsf{Rec}(\phi,-))$ is a secret sharing scheme for the access structure $\mathcal{A}_\phi$. Benaloh and Leichter [13] showed that access structures represented by monotone formulas admit efficient secret sharing schemes.

Lemma 6.11 ([13]). Let $\mathcal{A} := \{\mathcal{A}_\phi\}_{\phi\in\{0,1\}^*}$ be the family of access structures $\mathcal{A}_\phi := \{T \subseteq [n] : \phi(\chi_T) = 1\}$, where ϕ is a monotone formula on n variables and $\chi_T \in \{0,1\}^n$ denotes the characteristic vector of $T \subseteq [n]$. Then, $\mathcal{A}$ admits efficient secret sharing schemes.

Definition 6.12 (Minimum Monotone Satisfying Assignment; MMSA). For a monotone formula ϕ on n variables, the weight of an assignment $\alpha \in \{0,1\}^n$ is defined to be $\sum_{i=1}^n \alpha_i$. Let $\mathsf{MMSA}(\phi)$ denote the minimum weight of $\alpha \in \{0,1\}^n$ such that $\phi(\alpha) = 1$. Observe that $\mathsf{MMSA}(\phi) = w(\mathcal{A}_\phi)$ for the family $\mathcal{A}$ of access structures of Lemma 6.11.

Theorem 6.6. Let $c > 1$ be an arbitrary constant and $\tau : \mathbb{N}^3 \to \mathbb{N}$ be a function such that $\tau(n,m,t) \leq n^c \cdot m^{1-1/c} \cdot t^c$ for all large n, m, and $t \in \mathbb{N}$. Then, $\mathsf{Gap}_{\tau,\sigma}\mathsf{MINcKT}$ is NP-hard under one-query randomized polynomial-time reductions for some function $\sigma : \mathbb{N} \to \mathbb{N}$ such that $\sigma(n) = n^{1/(\log\log n)^{O(1)}}$.

Lemma 6.9. Let $(\mathsf{Share}, \mathsf{Rec})$ be a secret sharing scheme for an access structure $\mathcal{A}$ over $[n]$. Then, for every ℓ and $k \in \mathbb{N}$, it holds that

$$\Pr\left[\min_{T\notin\mathcal{A}} K(X \mid \mathsf{Share}(X)_T) \geq \ell - n - k\right] \geq 1 - 2^{-k},$$

where X is the uniform distribution over $\{0,1\}^\ell$ and the probability is taken over X as well as the internal randomness of Share.

Proof. Fix an arbitrary subset $T \notin \mathcal{A}$ of $[n]$. Let $y_T \in \mathrm{supp}(\mathsf{Share}(X)_T)$ be an outcome of $\mathsf{Share}(X)_T$. By Fact 3.2, we obtain

$$\Pr[K(X \mid y_T) < \ell - n - k] \leq 2^{-n-k}.$$

By the privacy of the secret sharing scheme, the random variables X and $\mathsf{Share}(X)_T$ are statistically independent. By averaging the inequality above over all $y_T \in \mathrm{supp}(\mathsf{Share}(X)_T)$, we obtain

$$\Pr[K(X \mid \mathsf{Share}(X)_T) < \ell - n - k] \leq 2^{-n-k}.$$

Finally, we take the union bound over all subsets $T \notin \mathcal{A}$ and conclude that

$$\Pr[\exists T \notin \mathcal{A}, K(X \mid \mathsf{Share}(X)_T) < \ell - n - k] \leq 2^{-k}.$$

□

It is known that MMSA is NP-hard to approximate:

Lemma 6.13 ([22,21]). For some function $g(n) = n^{1/(\log\log n)^{O(1)}}$, it is NP-hard to solve the promise problem $\mathsf{Gap}_g\mathsf{MMSA} = (\Pi_{\mathrm{Yes}}, \Pi_{\mathrm{No}})$ defined as follows:

$$\Pi_{\mathrm{Yes}} := \{(\phi, s) : \mathsf{MMSA}(\phi) \leq s\}, \qquad \Pi_{\mathrm{No}} := \{(\phi, s) : \mathsf{MMSA}(\phi) > s \cdot g(|\phi|)\},$$

where $|\phi|$ denotes the length of the binary string that represents ϕ.

Proof Sketch. Dinur and Safra [22] presented a polynomial-time reduction from any constraint satisfaction problem (CSP) to $\mathsf{Gap}_g\mathsf{MMSA}$ with the following property: Let ψ be an instance of the CSP. Let (ϕ, s) be the output of the reduction. Let D be the arity of ψ. If ψ is satisfiable, then $\mathsf{MMSA}(\phi) \leq s$. If there is no assignment satisfying a $1/(2(2Dg)^D)$ fraction of the constraints of ψ, then $\mathsf{MMSA}(\phi) > s \cdot g$, where g is an arbitrary parameter. Dinur et al. [21] showed that it is NP-hard to decide whether a given CSP with n variables and of arity $D = (\log\log n)^{O(1)}$ is satisfiable or every assignment satisfies at most a $1/\operatorname{poly}(n)$ fraction of the constraints. Combining these two hardness results, we conclude that $\mathsf{Gap}_g\mathsf{MMSA}$ is NP-hard, where $g \geq \operatorname{poly}(n)^{1/D} \cdot D^{-D} \geq n^{1/(\log\log n)^{O(1)}}$. □

Lemma 6.15 ([6,5]). Let $D = \{D_x\}_{x \in \{0,1\}^*}$ be a family of distributions sampled by circuits of size $\operatorname{poly}(|x|)$. Then, there exist a polynomial p and a constant $\delta > 0$ such that for every $x \in \{0,1\}^*$ and for every $D \geq p(|x|)$,

$$\frac{1}{D} \cdot K^t(y^1, \ldots, y^D) \leq H(D_x) + \frac{1}{2} \cdot D^{-\delta},$$

$$\frac{1}{D} \cdot K(y^1, \ldots, y^D) \geq H(D_x) - \frac{1}{2} \cdot D^{-\delta}$$

holds with probability at least $1 - 2^{-|x|}$ over a random choice of D independent samples $y^1, \ldots, y^D$ from D_x, where $t := p(|x|)$. This lemma immediately implies that the computational depth of independent samples from D_x is small.

Corollary 6.16. Under the same assumptions as Lemma 6.15, with probability at least $1 - 2^{-|x|}$ over a random choice of D independent samples $y^1, \ldots, y^D$ from D_x, it holds that

$$\mathrm{cd}^t(y^1, \ldots, y^D) \leq D^{1-\delta}.$$

Proof. It follows from Lemma 6.15 that

$$\mathrm{cd}^t(y^1, \ldots, y^D) = K^t(y^1, \ldots, y^D) - K(y^1, \ldots, y^D) \leq D^{1-\delta}.$$

□

6.2.3. A Generic Connection from Secret Sharing Schemes. We now formally state a generic reduction that reduces the problem of estimating the weight of access structures to $\mathsf{GapMINcKT}$.

Theorem 6.14. Let $\mathcal{A} = \{\mathcal{A}_\phi\}_{\phi\in\{0,1\}^*}$ be a family of access structures that admits efficient secret sharing schemes. Let $\tau : \mathbb{N}^3 \to \mathbb{N}$ be a function such that $t \le \tau(n, m, t) \le n^c \cdot m^{1-1/c} \cdot t^c$ for some constant $c > 1$. Then, there exists a randomized polynomial-time algorithm R that takes $\phi \in \{0,1\}^*$ as input and outputs $x, y \in \{0,1\}^*$ and $t, \rho \in \mathbb{N}$ such that

$$\frac{\rho}{8c} \cdot w(\mathcal{A}_\phi) + \log t' \le K^{t'}(x \mid y) \le K^t(x \mid y) \le \rho \cdot w(\mathcal{A}_\phi) - \mathrm{cd}^t(y)$$

holds with probability at least $1-o(1)$ over the internal randomness of R, where $t' := \tau(|x|, |y|, t)$. It is easy to observe that this reduction implies NP-hardness of $\mathsf{GapMINcKT}$.

Proof of Theorem 6.6. Let $\mathcal{A} = \{\mathcal{A}_\phi\}_\phi$ be the family of access structures for monotone formulas given in Lemma 6.11. Let R be the reduction of Theorem 6.14. Consider a reduction R' that reduces an instance (ϕ, s) of $\mathsf{Gap}_g\mathsf{MMSA}$ to an instance $(x, y, 1^t, 1^{s'})$ of $\mathsf{Gap}_\tau\mathsf{MINcKT}$ such that $(x, y, t, \rho) := R(\phi)$ and $s' := s\cdot\rho$. Let $(\Pi_{\mathrm{Yes}}, \Pi_{\mathrm{No}}) := \mathsf{Gap}_g\mathsf{MMSA}$. We claim below that the reduction R' reduces $\mathsf{Gap}_g\mathsf{MMSA}$ to $\mathsf{Gap}_{\tau,\sigma}\mathsf{MINcKT}$ for some σ, which implies NP-hardness of $\mathsf{Gap}_{\tau,\sigma}\mathsf{MINcKT}$ by Lemma 6.13. If $(\phi, s) \in \Pi_{\mathrm{Yes}}$, then $w(\mathcal{A}_\phi) = \mathsf{MMSA}(\phi) \le s$. By the properties of R, with high probability, it holds that

$$K^t(x \mid y) \le \rho \cdot w(\mathcal{A}_\phi) - \mathrm{cd}^t(y) \le s' - \mathrm{cd}^{t,t'}(y),$$

which implies that $(x, y, 1^t, 1^{s'})$ is a Yes instance of $\mathsf{Gap}_{\tau,\sigma}\mathsf{MINcKT}$. If $(\phi, s) \in \Pi_{\mathrm{No}}$, then $w(\mathcal{A}_\phi) = \mathsf{MMSA}(\phi) > g(|\phi|) \cdot s$. Also by the properties of R, with high probability, it holds that

$$K^{t'}(x \mid y) \ge \frac{\rho}{8c} \cdot w(\mathcal{A}_\phi) + \log t' > \frac{g(|\phi|)}{8c} \cdot s' + \log t',$$

which implies that $(x, y, 1^t, 1^{s'})$ is a No instance of $\mathsf{Gap}_{\tau,\sigma}\mathsf{MINcKT}$ for some $\sigma(|x|) := g(|\phi|)/(8c) \ge |\phi|^{1/(\log\log|\phi|)^{O(1)}} \ge |x|^{1/(\log\log|x|)^{O(1)}}$, where the last inequality follows from the fact that $|\phi| \ge |x|^{\Omega(1)}$. □

It remains to prove Theorem 6.14. To show that the additive error $\mathrm{cd}^t(y)$ is relatively small, we use the fact that for D independent random samples $y^1, \ldots, y^D$ from a distribution D samplable by polynomial-size circuits, the amortized time-bounded Kolmogorov complexity of $y^1, \ldots, y^D$ approaches the entropy of D asymptotically. We are now ready to prove Theorem 6.14.

Proof of Theorem 6.14. Fix an input $\phi \in \{0,1\}^*$. Let n be the number of parties in the access structure $\mathcal{A}_\phi$. Let $\lambda = O(\log|\phi|)$, t, and ℓ be parameters chosen later. We first define a family $D = \{D_\phi\}_{\phi\in\{0,1\}^*}$ of distributions by using the following sampling procedure: Choose $x \sim \{0,1\}^\ell$ uniformly at random. Let $(s_1, \ldots, s_n) := \mathsf{Share}(\phi, x)$ and let m be the length of each share; i.e., $m := |s_i|$. Pick $k_1, \ldots, k_n \sim \{0,1\}^\lambda$ randomly. We define a string $y \in \{0,1\}^{2^\lambda\cdot m}$, which we identify with a function $y : \{0,1\}^\lambda \to \{0,1\}^m$: For every $q \in \{0,1\}^\lambda$, let

$y(q) := s_i$ if $q = k_i$ for some $i \in [n]$ and let $y(q) := 0^m$ otherwise. Note that y is not well defined if $(k_i)_{i\in[n]}$ is not pairwise distinct; however, we will show that y is well defined with high probability. The output of the sampling procedure is defined as (x, y, k, s), where $k := (k_1, \ldots, k_n)$ and $s := (s_1, \ldots, s_n)$. The algorithm R operates as follows: For a given input $\phi \in \{0,1\}^*$, pick D independent samples $(x^1, y^1, k^1, s^1), \ldots, (x^D, y^D, k^D, s^D)$ from D_ϕ. Let $x := (x^1, \ldots, x^D)$, $y := (y^1, \ldots, y^D)$, and $\rho := 2\lambda D$. The output of R is defined to be (x, y, t, ρ). Below, we prove the correctness of R using a sequence of claims. Let $k := (k_i^d : i \in [n], d \in [D])$ and let $s := (s_i^d : i \in [n], d \in [D])$. We first observe that k is Kolmogorov-random with high probability:

Claim 6.17. With probability at least $1 - o(1)$, it holds that

$$K(k \mid s) \geq nD\lambda - \log n. \tag{16}$$

Proof. Since $k = (k_i^d)_{i\in[n],d\in[D]}$ is uniformly distributed over $(\{0,1\}^\lambda)^{nD}$ and independent of s, the claim follows from Fact 3.2. □

Claim 6.18. If Equation (16) holds, then $k = (k_i^d)_{i\in[n],d\in[D]}$ is pairwise distinct and hence y is well defined.

Proof. If there exists $(i, d) \neq (i', d')$ such that $k_i^d = k_{i'}^{d'}$, then k can be described by a program of size $nD\lambda - \lambda + O(\log nD)$. Choosing a large enough $\lambda = O(\log |\phi|)$, this implies that $K(k) \leq nD\lambda - \lambda + O(\log nD) < nD\lambda - \log n$, which contradicts Equation (16). □

Below, we assume that Equation (16) holds. We prove an upper bound of $K^t(x \mid y)$.

Claim 6.19. $K^t(x \mid y) \leq \rho \cdot w(\mathcal{A}_\phi) - \mathrm{cd}^t(y)$ holds with probability at least $1 - o(1)$.

Proof. Let $T \in \mathcal{A}_\phi$ be a minimum authorized set of parties such that $|T| = w(\mathcal{A}_\phi)$. We present an oracle program M^y that takes an index d and outputs $x^d \in \{0,1\}^\ell$ as follows: M^y takes $d \in [D]$ as input and T, $\{k_i^d : i \in T, d \in [D]\}$ and ϕ as hard-wired input, computes $\{s_i^d : i \in T\} = \{y^d(k_i^d) : i \in T\}$ by making queries to the oracle y, and outputs $\mathsf{Rec}(\phi, s_T^d)$, which is equal to x^d by the correctness of a secret sharing scheme and the assumption that $T \in \mathcal{A}_\phi$. The size of the oracle machine M is at most

$$O(|T| \cdot \log n) + |T| \cdot D \cdot \lambda + |\phi| + O(\log D) \leq 2\lambda D \cdot w(\mathcal{A}_\phi) - D^{1-\delta},$$

where $\delta > 0$ is the constant from Corollary 6.16 and this inequality follows by choosing sufficiently large $D \geq \mathrm{poly}(|\phi|)$. Applying Corollary 6.16 to the distribution of y, we obtain $\mathrm{cd}^t(y) \leq D^{1-\delta}$ with probability at least $1 - o(1)$; thus, $|M| \leq \rho \cdot w(\mathcal{A}_\phi) - \mathrm{cd}^t(y)$. The running time of M is at most $\mathrm{poly}(|\phi|)$, which is independent of λ. We choose t so that this running time is at most t. □

The remainder of the proof is devoted to proving the lower bound.

Claim 6.20. Let $\theta := \frac{\rho}{8c} \cdot w(\mathcal{A}_\phi) + \log t'$. Then, $K^{t'}(x \mid y) \geq \theta$ with probability at least $1 - o(1)$. We first clarify the condition of random variables under which the claim holds. Let $s_{[n]}^{[D]\setminus\{d\}} := (s_i^{d'} : i \in [n], d' \in [D] \setminus \{d\})$.

Claim 6.21. With probability at least $1 - o(1)$, it holds that for every unauthorized set $T \notin \mathcal{A}_\phi$ of parties and for every $d \in [D]$,

$$K(x^d \mid s_T^d, s_{[n]}^{[D]\setminus\{d\}}, k) \geq \ell - n - 2\log D. \tag{17}$$

Proof. Fix $d \in [D]$. Since the random variable x^d is independent of $s_{[n]}^{[D]\setminus\{d\}}$ and k, by Lemma 6.9,

$$\min_{T \notin \mathcal{A}_\phi} K(x^d \mid s_T^d, s_{[n]}^{[D]\setminus\{d\}}, k) \geq \ell - n - 2\log D$$

holds with probability at least $1 - 1/D^2$. The claim follows by taking a union bound over all $d \in [D]$. □

In what follows, we assume Equations (16) and (17) and prove $K^{t'}(x \mid y) \geq \theta$, which will complete the proof of Claim 6.20. Assume, by way of contradiction, that $K^{t'}(x \mid y) < \theta$. Let $M \in \{0,1\}^*$ be the description of an oracle program such that $|M| \leq \theta + O(\log \ell)$ and $M^y(d)$ outputs $x^d \in \{0,1\}^\ell$ in time $t'' := t'\ell$ for every $d \in [D]$; the machine M can be constructed from a program that witnesses $K^{t'}(x \mid y) < \theta$. For each $d \in [D]$, let $T(M,d)$ be the set of indices $i \in [n]$ such that some bit of $y^d(k_i^d) \in \{0,1\}^m$ is queried during the computation of $M^y(d)$.

Claim 6.22. Let M be an oracle program that runs in time t'' and let $\alpha := \sum_{d=1}^{D} |T(M,d)|$. Then,

$$K(k \mid s) \leq |M| + (nD - \alpha) \cdot \lambda + \alpha \cdot (\log t'' + \log nD) + O(\log nD).$$

Proof. By the definition of $T(M,d)$, for every $i \in T(M,d)$, there exists a time step $t_{i,d} \in [t'']$ such that M^y makes a query k_i^d to the oracle y on input $d \in [D]$. To describe k given s, consider the following program M': M' takes

$$\{(i, d, t_{i,d}) \in [n] \times [D] \times [t''] : i \in T(M,d)\},$$

$$\{k_i^d \in \{0,1\}^\lambda : (i,d) \in [n] \times [D],\ i \notin T(M,d)\},$$

and M as input and simulates M^y on input d for every $d \in [D]$. At the time step $t_{i,d}$ of the simulation of $M^y(d)$ for some $i \in T(M,d)$, M' reads the query k_i^d that $M^y(d)$ makes to the oracle y^d and answers the query with s_i^d. M' continues the simulation until the time step t'', at which point M' knows k_i^d for every $i \in T(M,d)$. Finally, M' outputs k. The input of M' can be encoded as a string of length $\alpha \cdot (\log t'' + \log nD) + (nD - \alpha) \cdot \lambda + |M| + O(\log nD)$. □

It follows from Claim 6.22 and Equation (16) that

$$nD\lambda - \log n \leq |M| + (nD - \alpha) \cdot \lambda + \alpha \cdot (\log t'' + \log nD) + O(\log nD),$$

which can be simplified to

$$(\lambda - \log t'' - \log nD) \cdot \alpha \leq |M| + O(\log nD). \tag{18}$$

Since $t''/\ell = t' = \tau(|x|, |y|, t) \leq |x|^c \cdot |y|^{1-1/c} \cdot t^c \leq 2^{(1-1/c)\cdot\lambda} \cdot (D \cdot |\phi| \cdot \ell \cdot m)^{O(c)}$, we may choose $\lambda = O(\log |\phi|)$ large enough so that $t'' \leq 2^{(1-1/2c)\cdot\lambda - \log nD}$. Then, by Equation (18), we obtain

$$\begin{aligned}\alpha &\leq \frac{2c}{\lambda} \cdot (|M| + O(\log nD)) \\ &< \frac{4c}{\lambda} \cdot (\theta - \log t') = \frac{\rho}{2\lambda} \cdot w(\mathcal{A}_\phi) = D \cdot w(\mathcal{A}_\phi),\end{aligned}$$

where the second inequality follows from $|M| + O(\log nD) \leq \theta + O(\log nD\ell t') < 2\theta$.[1] It follows that there exists $d \in [D]$ such that $|T(M, d)| \leq \alpha/D < w(\mathcal{A}_\phi)$, which implies that

$$T(M, d) \notin \mathcal{A}_\phi.$$

By Equation (17), we obtain

$$K(x^d \mid s^d_{T(M,d)}, s^{[D]\setminus\{d\}}_{[n]}, k) \geq \ell - n - 2\log D.$$

However, this contradicts the following claim.

Claim 6.23. Let M be an oracle program such that $M^y(d)$ outputs x^d for every $d \in [D]$. Then, for every $d \in [D]$, it holds that

$$K(x^d \mid s^d_{T(M,d)}, s^{[D]\setminus\{d\}}_{[n]}, k) \leq |M| + O(\log D).$$

Proof. Since $M^y(d)$ outputs x^d without making any query in $\{k^d_i : i \in [n] \setminus T(M, d)\}$, we can define another oracle $z = (z^1, \ldots, z^D)$ such that $z^{d'} := y^{d'}$ for every $d' \in [D] \setminus \{d\}$ and $z^d(k^d_i) := s^d_i$ for every $i \in T(M, d)$ and $z^d(k) := 0^m$ otherwise, so that there is no difference between y and z on inputs queried by $M^y(d)$; therefore, we obtain $x^d = M^y(d) = M^z(d)$. The claim follows by observing that the oracle z can be constructed from M, $d \in [D]$, $s^d_{T(M,d)}$, $s^{[D]\setminus\{d\}}_{[n]}$, and k. □

We conclude that $\ell - n - 2\log D \leq |M| + O(\log D) \leq O(\theta) \leq O(n\lambda D)$, which is a contradiction by letting $\ell \gg O(n\lambda D)$. This completes the proof of Claim 6.20. □

Remark 6.24. It is not hard to observe that our NP-hardness hardness reduction also proves the NP-hardness of McKTP [4] because the upper bound of Claim 6.19 also holds for $KT(x \mid y)$.

[1] We may assume without loss of generality that $w(\mathcal{A}_\phi) \geq 1$; then, we have $\theta \geq \Omega(\rho) \geq \Omega(D)$, which can be assumed to be larger than logarithmic terms.

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